BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `c:\Users\shini\hw5-jl3498-hw5`
```

```julia
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost $2000/day, tipping cost $50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost $1500/day, tipping cost $7/Mg, recycling cost $40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost $2500/day, tipping cost $60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are $1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
# Waste composition data
components = [
    ("Food Wastes",        15,   8,   0),
    ("Paper & Cardboard",  40,   7,  55),
    ("Plastics",            5,   5,  15),
    ("Textiles",            3,  10,  10),
    ("Rubber, Leather",     2,  15,   0),
    ("Wood",                5,   2,  30),
    ("Yard Wastes",        18,   2,  40),
    ("Glass",               4, 100,  60),
    ("Ferrous",             2, 100,  75),
    ("Aluminum",            2, 100,  80),
    ("Other Metal",         1, 100,  50),
    ("Miscellaneous",       3,  70,   0)
]
```

```julia
# compute totals
total_recycled = 0.0
total_ash = 0.0

for (name, mass_pct, ash_pct, rec_rate) in components
    frac = mass_pct/100
    recycle_frac = rec_rate/100
    ash_frac = ash_pct/100

    # amount recycled
    total_recycled += frac * recycle_frac

    # ash contribution (ash of total incoming waste)
    total_ash += frac * ash_frac
end

println("Overall recycling fraction = ", round(total_recycled; digits=4))
println("Overall ash fraction       = ", round(total_ash; digits=4))
```

```
Overall recycling fraction = 0.3775
Overall ash fraction       = 0.1641
```

This finds the overall recycling and ash fractions for each of the wastes produced through a for loop.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Decision Variables

- **Facility opening decisions**
  - $y_f \in \{0, 1\}$
    Indicates whether facility f is opened.
    * $y_f = 1$ if facility f (LF, MRF, or WTE) is opened and its fixed cost is incurred
    * $y_f = 0$ otherwise
- **City-to-facility waste flows**

- $x_{i,f} \geq 0$
  Amount of waste shipped from city i to facility f, where $i \in \{1, 2, 3\}$ and $f \in \{\text{LF}, \text{MRF}, \text{WTE}\}$.

- **MRF residual routing**
  - $r_{\text{MRF}\to\text{LF}} \geq 0$
    Amount of MRF residuals sent to the landfill.
  - $r_{\text{MRF}\to\text{WTE}} \geq 0$
    Amount of MRF residuals sent to the WTE facility.

- **WTE ash routing**
  - $a_{\text{WTE}\to\text{LF}} \geq 0$
    Amount of ash produced at the WTE facility that is sent to the landfill.

- **Facility throughputs** (auxiliary variables)
  - $m_{\text{MRF}} \geq 0$ — total mass entering the MRF
  - $m_{\text{WTE}} \geq 0$ — total mass entering the WTE
  - $m_{\text{LF}} \geq 0$ — total mass entering the landfill
  - $m_{\text{rec}} \geq 0$ — mass recycled at the MRF
  - $m_{\text{res}} \geq 0$ — mass of residuals produced by the MRF

These variables determine which facilities are opened, how waste flows through the network, how much material is recycled or landfilled, and the resulting system cost.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Objective Function — Total Daily System Cost

Parameters

- Facilities: $f \in \{\text{LF}, \text{MRF}, \text{WTE}\}$
- Fixed daily cost of opening facility f: F_f (\$/day)
  - $F_{LF} = 2000$, $F_{MRF} = 1500$, $F_{WTE} = 2500$

- Tipping (disposal) cost at facility f: c_f (\$/Mg)
  - $c_{LF} = 50$, $c_{MRF} = 7$, $c_{WTE} = 60$

- Recycling processing cost at MRF per Mg recycled: c_{rec} (\$/Mg)
  - $c_{\text{rec}} = 40$

- Transportation cost per Mg · km: τ (\$/Mg · km)
  - $\tau = 1.5$

Derivation — Cost Components

1. **Fixed facility costs:**
   Only incurred if the facility is opened.

$$\sum_{f \in \{\text{LF},\text{MRF},\text{WTE}\}} F_f \, y_f$$

2. **Tipping / disposal costs:**
   - Raw MSW shipped from cities to facilities:

$$\sum_{i=1}^{3} \sum_{f \in \{\text{LF},\text{MRF},\text{WTE}\}} c_f \, x_{i,f}$$

   - MRF residuals sent to landfill or WTE:

$$c_{\text{LF}} \, r_{\text{MRF}\to\text{LF}} + c_{\text{WTE}} \, r_{\text{MRF}\to\text{WTE}}$$

   - WTE ash sent to landfill:

$$c_{\text{LF}} \, a_{\text{WTE}\to\text{LF}}$$

3. **MRF recycling processing cost:**

$$c_{\text{rec}} \, m_{\text{rec}} = c_{\text{rec}} \, R_{\text{MRF}} \, m_{\text{MRF}}$$

4. **Transportation costs:**

   - Cities $\to$ facilities:

$$\tau \sum_{i=1}^{3} \sum_{f \in \{\mathrm{LF},\mathrm{MRF},\mathrm{WTE}\}} x_{i,f}\, d_{i,f}$$

   - Inter-facility flows (MRF residuals, WTE ash):

$$\tau \left(r_{\mathrm{MRF}\to\mathrm{LF}}\, d_{\mathrm{MRF,LF}} + r_{\mathrm{MRF}\to\mathrm{WTE}}\, d_{\mathrm{MRF,WTE}} + a_{\mathrm{WTE}\to\mathrm{LF}}\, d_{\mathrm{WTE,LF}} \right)$$

Full Objective

Minimize total daily system cost Z:

$$\begin{aligned}
\min Z = \; & \underbrace{\sum_f F_f\, y_f}_{\text{fixed facility costs}} + \underbrace{\sum_{i=1}^{3} \sum_f c_f\, x_{i,f}}_{\text{tipping of raw MSW}} + \underbrace{c_{\mathrm{rec}}\, R_{\mathrm{MRF}}\, m_{\mathrm{MRF}}}_{\text{MRF recycling cost}} \\
& + \underbrace{c_{\mathrm{LF}}\, r_{\mathrm{MRF}\to\mathrm{LF}} + c_{\mathrm{WTE}}\, r_{\mathrm{MRF}\to\mathrm{WTE}}}_{\text{tipping of MRF residuals}} + \underbrace{c_{\mathrm{LF}}\, a_{\mathrm{WTE}\to\mathrm{LF}}}_{\text{tipping of WTE ash}} \\
& + \underbrace{\tau \sum_{i=1}^{3} \sum_f x_{i,f}\, d_{i,f}}_{\text{transport: cities} \to \text{facilities}} + \underbrace{\tau \left(r_{\mathrm{MRF}\to\mathrm{LF}}\, d_{\mathrm{MRF,LF}} + r_{\mathrm{MRF}\to\mathrm{WTE}}\, d_{\mathrm{MRF,WTE}} + a_{\mathrm{WTE}\to\mathrm{LF}}\, d_{\mathrm{WTE,LF}} \right)}_{\text{transport: inter-facility / ash}}
\end{aligned}$$

Auxiliary Relationships

- Total mass entering MRF: $m_{\mathrm{MRF}} = \sum_{i=1}^{3} x_{i,\mathrm{MRF}}$
- Total mass recycled at MRF: $m_{\mathrm{rec}} = R_{\mathrm{MRF}}\, m_{\mathrm{MRF}}$
- Residual mass leaving MRF: $m_{\mathrm{res}} = (1 - R_{\mathrm{MRF}})\, m_{\mathrm{MRF}}$
- Residual allocation: $r_{\mathrm{MRF}\to\mathrm{LF}} + r_{\mathrm{MRF}\to\mathrm{WTE}} = m_{\mathrm{res}}$
- Ash from WTE: $a_{\mathrm{WTE}\to\mathrm{LF}} = \alpha_{\mathrm{raw}} \sum_{i=1}^{3} x_{i,\mathrm{WTE}} + \alpha_{\mathrm{res}}\, r_{\mathrm{MRF}\to\mathrm{WTE}}$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Constraints — Mass Balances, Capacities, and Linking

1) City supply balances

All waste produced by each city must be assigned to a facility:

$$\sum_{f \in \{\mathrm{LF}, \mathrm{MRF}, \mathrm{WTE}\}} x_{i,f} = S_i, \quad i = 1, 2, 3$$

Ensures that all waste generated by a city is sent to some facility (mass conservation).

---

2) Facility throughput definitions

Total mass entering each facility:

- **MRF:**

$$m_{\mathrm{MRF}} = \sum_{i=1}^{3} x_{i,\mathrm{MRF}}$$

- **WTE:**

$$m_{\mathrm{WTE}} = \sum_{i=1}^{3} x_{i,\mathrm{WTE}} + r_{\mathrm{MRF} \to \mathrm{WTE}}$$

- **Landfill:**

$$m_{\mathrm{LF}} = \sum_{i=1}^{3} x_{i,\mathrm{LF}} + r_{\mathrm{MRF} \to \mathrm{LF}} + a_{\mathrm{WTE} \to \mathrm{LF}}$$

Defines the total mass processed by each facility, which will be used for capacity constraints and cost calculation.

---

3) MRF recycling and residuals

- **Recycled mass:**

$$m_{\mathrm{rec}} = R_{\mathrm{MRF}}\, m_{\mathrm{MRF}}$$

- **Residual (non-recycled) mass:**

$$m_{\mathrm{res}} = (1 - R_{\mathrm{MRF}})\, m_{\mathrm{MRF}}$$

- **Residual allocation:**

$$r_{\mathrm{MRF}\to\mathrm{LF}} + r_{\mathrm{MRF}\to\mathrm{WTE}} = m_{\mathrm{res}}$$

with

$$r_{\mathrm{MRF}\to\mathrm{LF}} \geq 0, \quad r_{\mathrm{MRF}\to\mathrm{WTE}} \geq 0$$

The MRF splits incoming mass into recycled material and residuals; residuals must be assigned to a downstream facility.

---

4) WTE ash generation and routing

All ash produced at the WTE is sent to the landfill:

$$a_{\mathrm{WTE}\to\mathrm{LF}} = \alpha_{\mathrm{raw}} \sum_{i=1}^{3} x_{i,\mathrm{WTE}} + \alpha_{\mathrm{res}}\, r_{\mathrm{MRF}\to\mathrm{WTE}}$$

Ash fractions differ for raw MSW and residuals; the formula calculates total ash directed to the landfill.

---

5) Facility capacity constraints

Each facility cannot exceed its processing capacity, and flows are only allowed if the facility is opened:

- **Landfill:**

$$\sum_{i=1}^{3} x_{i,\mathrm{LF}} + r_{\mathrm{MRF}\to\mathrm{LF}} + a_{\mathrm{WTE}\to\mathrm{LF}} \leq C_{\mathrm{LF}}\, y_{\mathrm{LF}}$$

- **MRF:**

$$\sum_{i=1}^{3} x_{i,\mathrm{MRF}} \leq C_{\mathrm{MRF}}\, y_{\mathrm{MRF}}$$

- **WTE:**

$$\sum_{i=1}^{3} x_{i,\mathrm{WTE}} + r_{\mathrm{MRF}\to\mathrm{WTE}} \leq C_{\mathrm{WTE}}\, y_{\mathrm{WTE}}$$

Standard capacity constraints. If $y_f = 0$, the RHS is zero, preventing flow to a closed facility.

---

6) Linking constraints for flows

To prevent shipments to closed facilities:

$$\sum_{i=1}^{3} x_{i,f} \leq C_f\, y_f, \quad f \in \{\mathrm{LF}, \mathrm{MRF}, \mathrm{WTE}\}$$

Residuals and ash cannot be routed to closed facilities:

$$r_{\mathrm{MRF}\to\mathrm{LF}} \leq C_{\mathrm{LF}}\, y_{\mathrm{LF}}, \quad r_{\mathrm{MRF}\to\mathrm{WTE}} \leq C_{\mathrm{WTE}}\, y_{\mathrm{WTE}}, \quad a_{\mathrm{WTE}\to\mathrm{LF}} \leq C_{\mathrm{LF}}\, y_{\mathrm{LF}}$$

Big-M constraints ensure flows only occur if the facility is open; using C_f as M_f keeps the bound tight.

---

7) Nonnegativity

$$x_{i,f} \geq 0, \quad r_{\mathrm{MRF}\to\mathrm{LF}} \geq 0, \quad r_{\mathrm{MRF}\to\mathrm{WTE}} \geq 0, \quad a_{\mathrm{WTE}\to\mathrm{LF}} \geq 0$$

$$y_f \in \{0,1\}, \quad f \in \{\mathrm{LF}, \mathrm{MRF}, \mathrm{WTE}\}$$

Waste flows cannot be negative; facility opening decisions are binary.

---

8) policy constraints

- Minimum recycling target:

$$m_{\mathrm{rec}} = R_{\mathrm{MRF}}\, m_{\mathrm{MRF}} \geq T$$

- Force facility to open:

$$y_{\mathrm{LF}} = 1$$

- Mutually exclusive facility selection:

$$\sum_{f \in \mathrm{MRF\ candidates}} y_f \leq 1$$

Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

```julia
# Given data
cities = 1:3
facilities = [:LF, :MRF, :WTE]
# City waste (Mg/day)
S = Dict(1=>100, 2=>90, 3=>120)
# Facility capacities (Mg/day)
C = Dict(:LF=>200, :MRF=>350, :WTE=>210)
# Fixed costs ($/day)
F = Dict(:LF=>2000, :MRF=>1500, :WTE=>2500)
# Tipping costs ($/Mg)
```

```julia
c = Dict(:LF=>50, :MRF=>7, :WTE=>60)
# Recycling processing cost ($/Mg recycled)
c_rec = 40
# Transportation cost ($/Mg·km)
t = 1.5

# Distances (km) from cities to facilities
d = Dict(
    (1,:LF)=>5, (1,:MRF)=>30, (1,:WTE)=>15,
    (2,:LF)=>15, (2,:MRF)=>25, (2,:WTE)=>10,
    (3,:LF)=>13, (3,:MRF)=>45, (3,:WTE)=>20)
# Inter-facility distances
d_inter = Dict(
    (:MRF,:LF)=>32, (:MRF,:WTE)=>15, (:WTE,:LF)=>18)
# MRF recycling fraction
R_MRF = 0.3775
# WTE ash fractions
a_raw = 0.1641
a_res = 0.14


# Build model
model = Model(HiGHS.Optimizer)
# Decision variables
@variable(model, y[f in facilities], Bin)
@variable(model, x[i in cities, f in facilities] >= 0)
@variable(model, r_MRF_LF >= 0)
@variable(model, r_MRF_WTE >= 0)
@variable(model, a_WTE_LF >= 0)
# Auxiliary variables
@expression(model, m_MRF, sum(x[i,:MRF] for i in cities))
@expression(model, m_WTE, sum(x[i,:WTE] for i in cities) + r_MRF_WTE)
@expression(model, m_LF, sum(x[i,:LF] for i in cities) + r_MRF_LF + a_WTE_LF)
@expression(model, m_rec, R_MRF*m_MRF)
@expression(model, m_res, (1-R_MRF)*m_MRF)


# Constraints
# 1) City supply balances
for i in cities
    @constraint(model, sum(x[i,f] for f in facilities) == S[i])
end
```

```julia
# 2) MRF residual allocation
@constraint(model, r_MRF_LF + r_MRF_WTE == m_res)
# 3) WTE ash
@constraint(model, a_WTE_LF == a_raw*sum(x[i,:WTE] for i in cities) + a_res*r_MRF_WTE)
# 4) Capacity constraints
@constraint(model, m_LF <= C[:LF]*y[:LF])
@constraint(model, m_MRF <= C[:MRF]*y[:MRF])
@constraint(model, m_WTE <= C[:WTE]*y[:WTE])
# 5) Linking constraints (prevent flow to closed facilities)
for f in facilities
    @constraint(model, sum(x[i,f] for i in cities) <= C[f]*y[f])
end
@constraint(model, r_MRF_LF <= C[:LF]*y[:LF])
@constraint(model, r_MRF_WTE <= C[:WTE]*y[:WTE])
@constraint(model, a_WTE_LF <= C[:LF]*y[:LF])


# Objective function
@expression(model, obj,
    # Fixed costs
    sum(F[f]*y[f] for f in facilities) +
    # Tipping raw MSW
    sum(c[f]*x[i,f] for i in cities, f in facilities) +
    # Recycling processing
    c_rec * m_rec +
    # MRF residual tipping
    c[:LF]*r_MRF_LF + c[:WTE]*r_MRF_WTE +
    # WTE ash tipping
    c[:LF]*a_WTE_LF +
    # Transportation: cities to facilities
    t*sum(x[i,f]*d[(i,f)] for i in cities, f in facilities) +
    # Transportation: inter-facility / ash
    t*(r_MRF_LF*d_inter[:MRF,:LF] + r_MRF_WTE*d_inter[:MRF,:WTE] + a_WTE_LF*d_inter[:WTE,:LF]
)

@objective(model, Min, obj)

# Solve
optimize!(model)

# Results
status = termination_status(model)
```

```julia
println("Solver status: ", status)

if status == MOI.OPTIMAL
    println("Optimal total daily cost: \$", objective_value(model))
    println("Facility openings:")
    for f in facilities
        println("  ", f, ": ", value(y[f]))
    end
    println("City-to-facility flows:")
    for i in cities, f in facilities
        println("  City ", i, " → ", f, ": ", value(x[i,f]))
    end
    println("MRF residuals → LF: ", value(r_MRF_LF))
    println("MRF residuals → WTE: ", value(r_MRF_WTE))
    println("WTE ash → LF: ", value(a_WTE_LF))
end
```

```
Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 14 rows; 15 cols; 52 nonzeros; 3 integer variables (3 binary)
Coefficient ranges:
  Matrix  [1e-01, 4e+02]
  Cost    [6e+01, 2e+03]
  Bound   [1e+00, 1e+00]
  RHS     [9e+01, 1e+02]
Presolving model
14 rows, 15 cols, 52 nonzeros  0s
10 rows, 13 cols, 40 nonzeros  0s
Presolve reductions: rows 10(-4); columns 13(-2); nonzeros 40(-12)

Solving MIP model with:
   10 rows
   13 cols (2 binary, 0 integer, 0 implied int., 11 continuous, 0 domain fixed)
   40 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized rou
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS soluti
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial

        Nodes      |    B&B Tree     |            Objective Bounds              |  Dynamic Co
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap |   Cuts   In
```

```
 J       0       0         0   0.00%   -inf             40650.317587      Large        0
 S       0       0         0   0.00%   2000             27855.482115     92.82%        0

50.0% inactive integer columns, restarting
Model after restart has 1 row, 3 cols (0 bin., 0 int., 0 impl., 3 cont., 0 dom.fix.), and 3 r

         0       0         0   0.00%   26922.085548     27855.482115      3.35%        0
 T       0       0         0   0.00%   26922.085548     27855.482115      3.35%        0
         1       0         1 100.00%   27855.482115     27855.482115      0.00%        0

Solving report
  Status            Optimal
  Primal bound      27855.4821151
  Dual bound        27855.4821151
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.00779575166961
  Solution status   feasible
                    27855.4821151 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.03
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     6
                    0 (strong br.)
                    0 (separation)
                    0 (heuristics)
Solver status: OPTIMAL
Optimal total daily cost: $27855.48211508554
Facility openings:
  LF: 1.0
  MRF: 0.0
  WTE: 1.0
City-to-facility flows:
  City 1 → LF: 100.0
  City 1 → MRF: 0.0
  City 1 → WTE: 0.0
  City 2 → LF: 0.0
  City 2 → MRF: 0.0
  City 2 → WTE: 90.0
  City 3 → LF: 78.40531164014834
```

```
  City 3 → MRF: 0.0
  City 3 → WTE: 41.594688359851666
MRF residuals → LF: 0.0
MRF residuals → WTE: 0.0
WTE ash → LF: 21.59468835985166
```

This code uses the Decision variables, Auxiliary variables, Constraints, and Objective function from the previous question through the optimization model.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

City1 (100) —–> LF (open)

City2 (90) —–> WTE (open) –(ash 21.59)–> LF

City3 (41.59)-> WTE

City3 (79.04) -> LF

MRF : CLOSED (no flows)

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100$MW. In period 2, the demand is projected to be $d_2 = 1200$MW, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

```
                  ROOT (time 0)
                     |
                     |  (deterministic period 1)
                     v
         Node t=1: d1 = 1100 MW, solar CF=0.90, wind CF=0.45
```

```
                              |
         ----------------------------------------------------------
         |                          |                        |
         |                          |                        |
 (demand branch)            (demand branch)           (conceptually same
 d2=1200 (0.75)             d2=1500 (0.25)           single node at t=1)
         |                          |
  -------------------          -------------------------
  |         |       |             |       |         |
 (renew) (renew) (renew)      (renew)  (renew)    (renew)
  0.70    0.30               0.70         0.30
  |        |                   |            |
  |        |                   |            |
Scen A    Scen B            Scen C       Scen D

 (1200,    (1200,            (1500,      (1500,

 0.95/0.40) 0.75/0.50)        0.95/0.40) 0.75/0.50)

 prob=0.525  prob=0.225       prob=0.175  prob=0.075
```

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

```
gen = CSV.read("data/generators.csv", DataFrame)

# Define sets and parameters
G = Vector{String}(gen.Plant)
Pmin = Dict(gen.Plant .=> gen.Pmin)
Pmax = Dict(gen.Plant .=> gen.Pmax)
VarCost = Dict(gen.Plant .=> gen.VarCost)
Ramp = Dict(gen.Plant .=> gen.Ramp)

# Identify solar and wind units
solar_units = gen.Plant[gen.Resource .== "solar"]
wind_units  = gen.Plant[gen.Resource .== "wind"]

# SCENARIOS AND DATA
T = 1:2                                  # time periods
S = 1:2                                  # 2 end scenarios for period 2
```

```
# Scenario probabilities
pi = Dict(1 => 0.75,  # normal demand, normal renewables
          2 => 0.25)  # high demand, stressed renewables

# Demands
d = Dict((1,1)=>1100.0, (1,2)=>1100.0,  # period 1 deterministic
         (2,1)=>1200.0, (2,2)=>1500.0)  # period 2 uncertain

# Solar and wind capacity factors
solar_cf = Dict((1,1)=>0.90, (1,2)=>0.90,
                (2,1)=>0.95, (2,2)=>0.75)

wind_cf  = Dict((1,1)=>0.45, (1,2)=>0.45,
                (2,1)=>0.40, (2,2)=>0.50)


model = Model(HiGHS.Optimizer)

# DECISION VARIABLES
# Generation per generator, period, and scenario
p = @variable(model, [g in G, t in T, s in S])

# CONSTRAINTS
#Power limits: Pmin <= p <= Pmax
@constraint(model, [g in G, t in T, s in S],
    Pmin[g] <= p[g,t,s] <= Pmax[g])

#Renewable availability (capacity factor)
@constraint(model, [g in solar_units, t in T, s in S], p[g,t,s] <= Pmax[g] * solar_cf[(t,s)]
@constraint(model, [g in wind_units, t in T, s in S], p[g,t,s] <= Pmax[g] * wind_cf[(t,s)])

#Ramping constraints period 1 to 2
@constraint(model, [g in G, s in S], p[g,2,s] - p[g,1,s] <= Ramp[g])
@constraint(model, [g in G, s in S], p[g,1,s] - p[g,2,s] <= Ramp[g])

#Supply-demand balance for each period, each scenario
@constraint(model, [t in T, s in S], sum(p[g,t,s] for g in G) == d[(t,s)])

#Minimize expected generation cost
@objective(model, Min, sum(pi[s] * sum(VarCost[g] * p[g,t,s] for g in G) for t in T, s in S)

println("Stochastic economic dispatch model formulated successfully.")
```

```
Warning: thread = 1 warning: parsed expected 6 columns, but didn't reach end of line around
@ CSV C:\Users\shini\.julia\packages\CSV\XLcqT\src\file.jl:593
Warning: thread = 1 warning: only found 6 / 7 columns around data row: 4. Filling remaining
@ CSV C:\Users\shini\.julia\packages\CSV\XLcqT\src\file.jl:592
Warning: thread = 1 warning: only found 6 / 7 columns around data row: 5. Filling remaining
@ CSV C:\Users\shini\.julia\packages\CSV\XLcqT\src\file.jl:592
Warning: thread = 1 warning: only found 6 / 7 columns around data row: 6. Filling remaining
@ CSV C:\Users\shini\.julia\packages\CSV\XLcqT\src\file.jl:592
```

```
Stochastic economic dispatch model formulated successfully.
```

References

List any external references consulted, including classmates.